DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

MODULE II QUESTION BANK

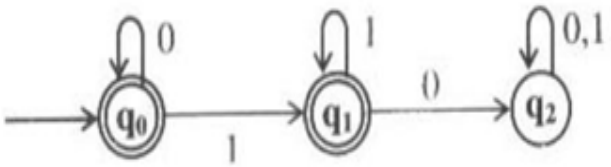
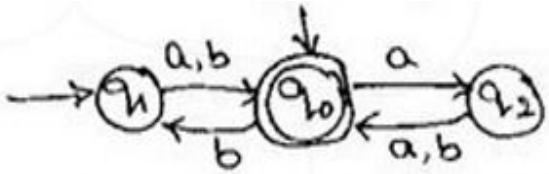
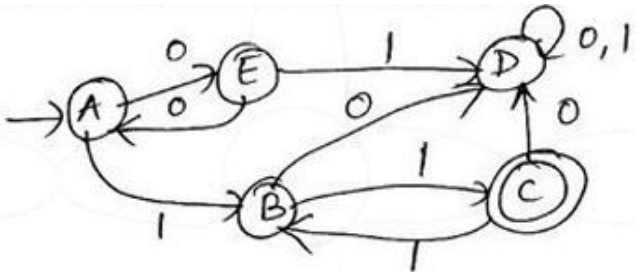
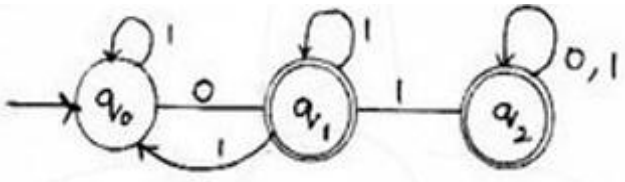
Course Title:	**Theory of Computation**	**Course Code:**	**BCS503**
Semester:	**V**	**Academic Year:**	**2026-2027**
Batch:	**2024-2028**		

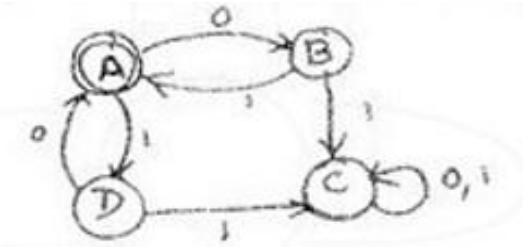
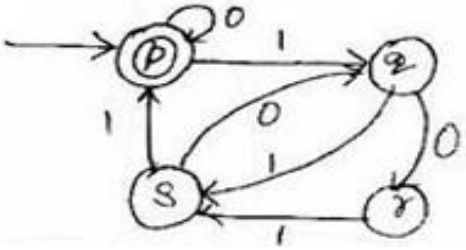
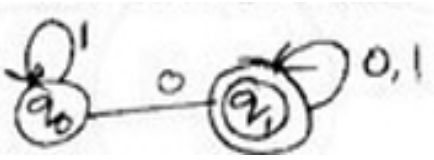
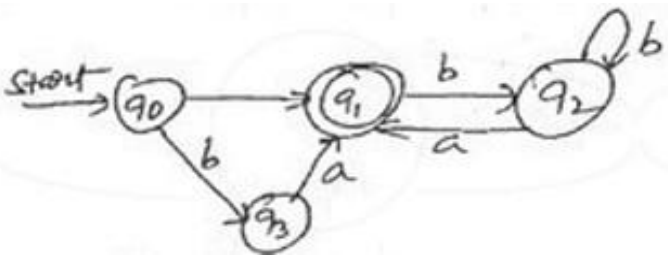
Module II

Regular Expressions and Languages: Regular Expressions, Finite Automata and Regular Expressions, Applications of Regular Expressions, Proving Languages not to be Regular, Closure Properties of Regular Languages.

TextBook	1
Chapter	Textbook 1: 3.1, 3.2 (Except 3.2.1), 3.3, 4.1, 4.2
RBT	1, 2, 3

Sl. No.	**Questions**	**COs**	**RBT Level**
1	List applications of RE. What are the notations used in UNIX Operation system? List few Regular expressions with its UNIX notations.	1	2
2	Obtain an ∈-NFA for the Regular Expression i) (a+b)* bb (a+b)* ii) ab(a+b)*	2	3
3	Construct ∈-NFA for the following Regular Expression i) (0+11)0*1 ii) 0+01*	2	3
4	a. Obtain NFA which accepts strings of a's and b's starting with the string ab. b. Obtain NFA for the Regular Expression (a+b)*aa(a+b)*	2	3
5	Define Pumping Lemma. State and prove pumping lemma Theorem.	2	2
6	a. Prove that below language is not a regular Language. i) L ={ $a^i b^j$ | i > j } ii) L={a^p|p is prime} b. Show that, L= {WW^R | W Є {a,b) *}} is not regular	2	3
7	Develop Regular expressions for the following Languages on Σ={a, b} i) L={$a^n b^m$|n+m is even} ii) L={$a^{2n} b^{2m}$|n>=0,m>=0}	2	3

	iii) L={w:n_a(w) mod 3=0 where w Є(a+b)*} iv) Accept strings of a's and b's whose fifth symbol from the right end is a. v) Accept strings of a's and b's containing not more than 3 a's.		
8	Define Regular expression. Obtain regular expression for the following languages: i) Representing for strings of a's and b's having odd length. ii) To accept strings of a's and b's such that third symbol from the right is a and fourth symbol from the right is b. iii) L={0^n1^m|m>=1,n>=1,mn>=3}	2	3
9	Obtain Regular Expression for the following languages L={$a^nb^mc^p$|n<=4,m>=2,p<=2}	2	3
10	Find Regular language accepted by the following FA by eliminating states? 	2	3
11	Show that regular languages are closed under, i) Union, Concatenation and Kleens star ii) Intersection and Difference iii) Complement	2	2
12	Show that the regular languages are closed under Homomorphism.	2	2
13	Convert the following automata to a regular expression. 	2	3
14	Build a regular expression for the given FSM. 	2	3
15	Obtain the Regular expression for the following FSM. 	2	3

16	Build a regular expression for the given FSM. 	2	3
17	Write the equivalent Regular Expression for the given Finite State Machine. 	2	3
18	Obtain a RE for the finite automata. 	2	3
19	Construct a FA for the regular expression 10+(0+11)0*1	2	3
20	List the closure properties of Regular Languages.	2	3
21	Write DFSM to accept intersection of Languages L1=(a+b)*a and L2=(a+b)*b	2	3
22	Construct an NFSM which accepts the regular expression (a+b)*abb	2	3
23	Let L be the language accepted by the following finite state machine.  Indicate for each of the following regular expeession, whether it correctly describes L: i) (aUba)bb*a ii) (ϵUb)a(bb*a)* iii) baUab*a iv) baUab*aUa	2	3

24	Write the Regular Expression for the following language. i) {w Є(a,b)* with almost one a} ii) {w Є(a,b)* does not end with ba} iii) {w Є(0,1)* has substring 001} iv) {w Є(0,1)* |W| is even	2	3
25	Show that for every RE there is an equivalent FSM	2	3

Faculty Incharge
(Dr. S. Mercy)

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**